

7EZ2_hs_9

Tetrahymena ribozyme. Helices are labeled with a large font size, basepairs are labeled outside the circle.

Note kissing hairpins P2 with P5c and P2.1 with P9.1a.

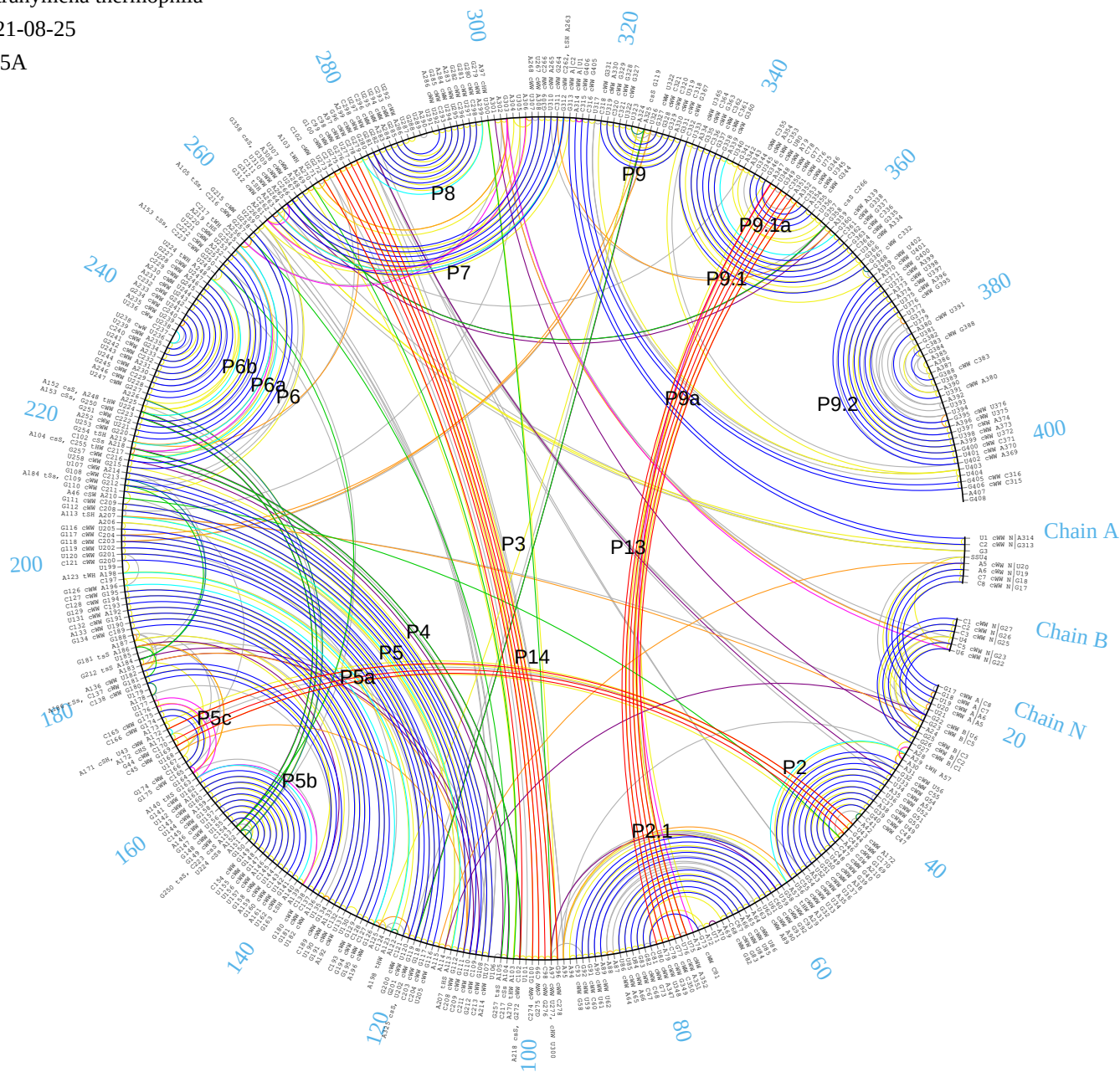
Holo L-16 Scal Tetrahymena ribozyme

ELECTRON MICROSCOPY

Tetrahymena thermophila

2021-08-25

3.05A



- 119 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 15 long-range Watson-Crick basepairs
- 13 nested non-Watson-Crick basepairs (Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 13 long-range non-Watson-Crick basepairs
- 430 base stacking interactions
- 13 base-phosphate interactions
- 35 base-ribose interactions
- 18 sugar-ribose interactions
- 8 oxygen stacking interactions
- 124 near basepairs or other interactions

Chain A: Holo L-16 Scal Tetrahymena ribozyme S1
Chain B: Holo L-16 Scal Tetrahymena ribozyme S2
Chain N: Group I catalytic intron

7K00

Complete E. coli ribosome. Red arcs show two tRNAs interacting with mRNA, other arcs show where they touch the SSU and LSU.

Helix numbers are shown; for example, SSU helix 44 interacts with LSU Helix 79, and 5S helix 4 interacts with LSU Helix 38.

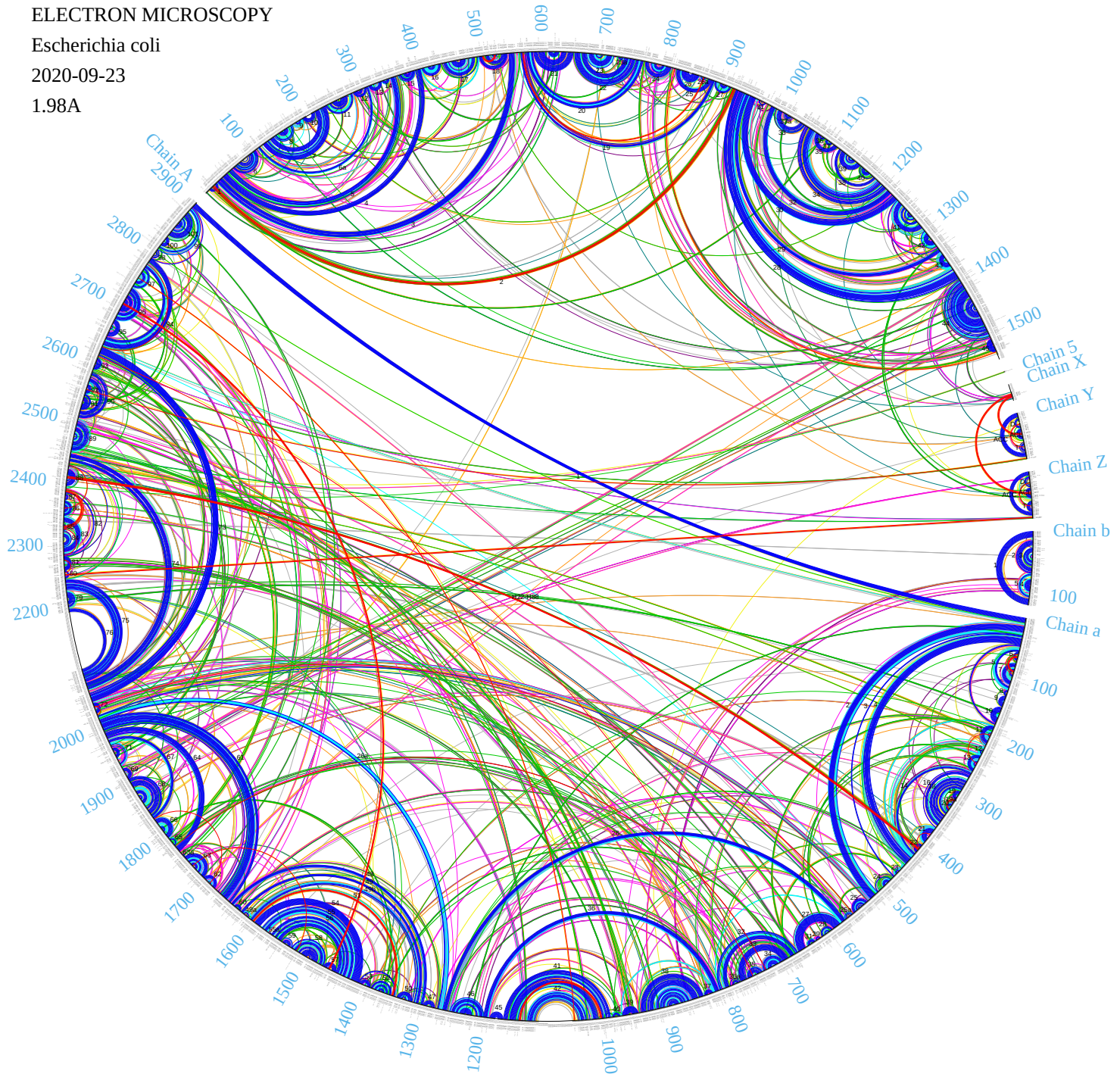
Structure of the Bacterial Ribosome at 2 Angstrom Resolution

ELECTRON MICROSCOPY

Escherichia coli

2020-09-23

1.98A



- 1257 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 57 long-range Watson-Crick basepairs
- 447 nested non-Watson-Crick basepairs (Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 272 long-range non-Watson-Crick basepairs
- 4910 base stacking interactions
- 582 base-phosphate interactions
- 612 base-ribose interactions
- 238 sugar-ribose interactions
- 251 oxygen stacking interactions
- 1223 near basepairs or other interactions

- Chain b: 5S ribosomal RNA
- Chain a: Large subunit ribosomal RNA
- Chain A: Small subunit ribosomal RNA
- Chain 5: E-site tRNA
- Chain X: mRNA
- Chain Y: Transfer RNA
- Chain Z: Transfer RNA

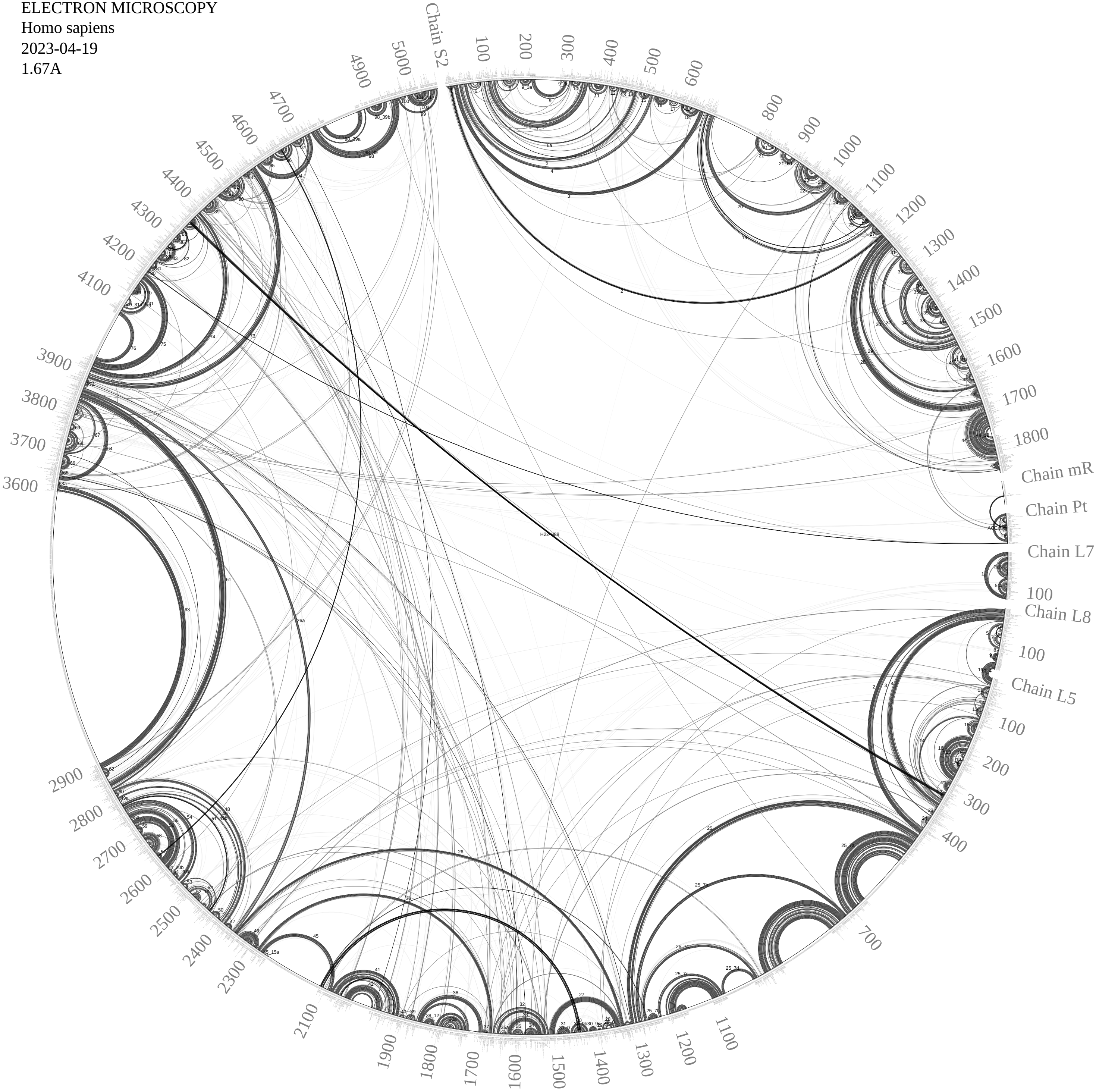
8GLP_grayscale_hide_stacking,near_dim_bph,br,sr,so_text_basepair,bph,br,sr,so
Human 5.8S rRNA and large subunit. Large page size. Nucleotides in the LSU that do not have xyz coordinates are listed with their chain position numbers.
Interactions shown in grayscale. Base-backbone interaction arcs are dimmed; base stacking and near interactions are hidden to allow more focus on basepairs.
mRNA decoding in human is kinetically and structurally distinct from bacteria (Consensus LSU focused refined structure)

ELECTRON MICROSCOPY

Homo sapiens

2023-04-19

1.67A



- 1499 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 58 long-range Watson-Crick basepairs
- 471 nested non-Watson-Crick basepairs (Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 227 long-range non-Watson-Crick basepairs
- 587 base-phosphate interactions
- 649 base-ribose interactions
- 223 sugar-ribose interactions
- 267 oxygen stacking interactions

- Chain L7: 5S ribosomal RNA
- Chain L8: 5.8S ribosomal RNA
- Chain L5: Large subunit ribosomal RNA
- Chain S2: Small subunit ribosomal RNA
- Chain mR: mRNA
- Chain Pt: Transfer RNA

7JQQ_K_L_M_N_O_wong_text_basepair,bph,br,so

Bacteriophage Phi-29 viral genome packaging motor assembly. Wong color scheme to help with color blindness. Five-fold symmetry with five pseudoknots. Basepairs, base-backbone and oxygen stackings listed around the circle.

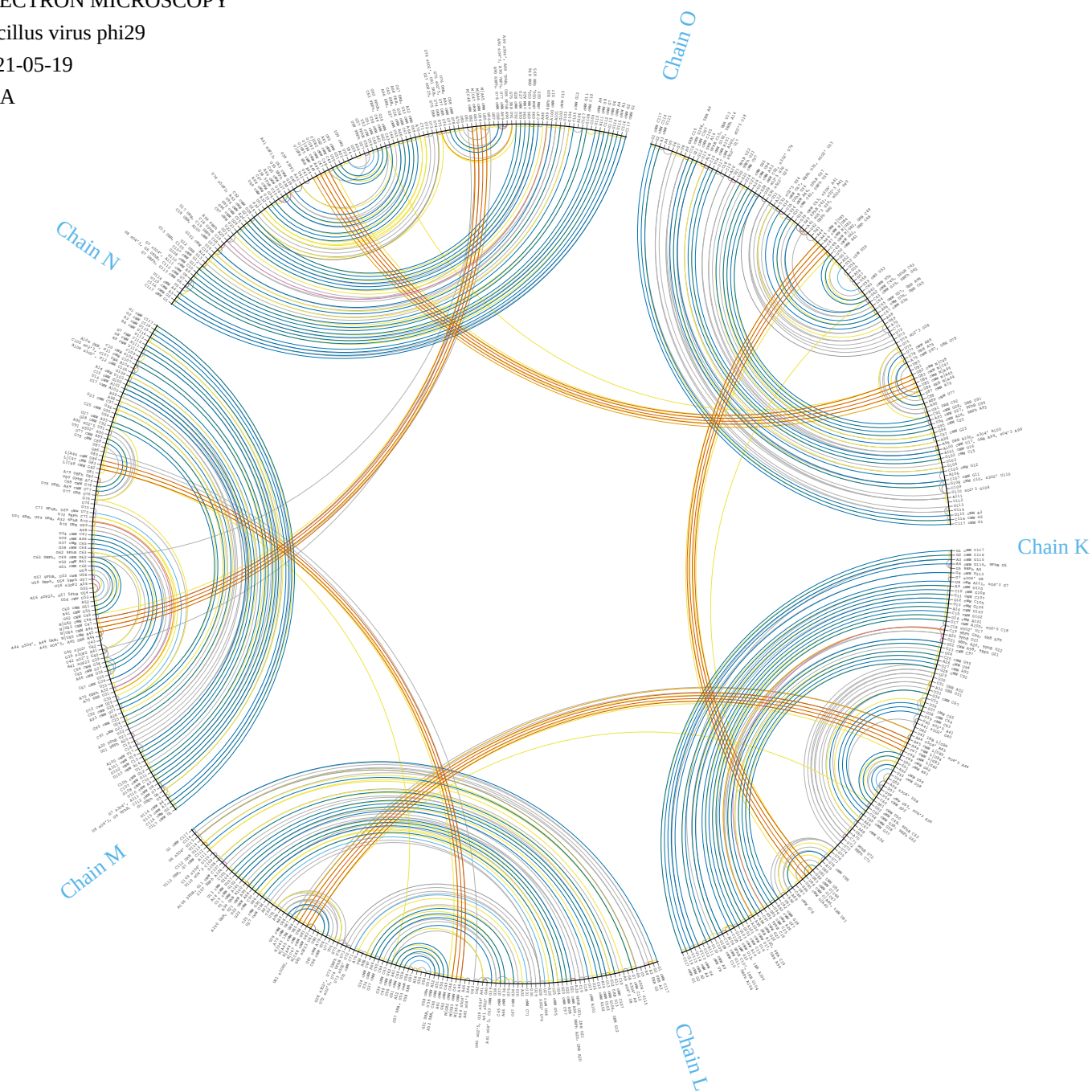
The bacteriophage Phi-29 viral genome packaging motor assembly

ELECTRON MICROSCOPY

Bacillus virus phi29

2021-05-19

4.1A



- 129 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 19 long-range Watson-Crick basepairs
- 9 nested non-Watson-Crick basepairs (Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 0 long-range non-Watson-Crick basepairs
- 450 base stacking interactions
- 33 base-phosphate interactions
- 33 base-ribose interactions
- 0 sugar-ribose interactions
- 34 oxygen stacking interactions
- 270 near basepairs or other interactions

Chain K: Bacteriophage pRNA
Chain L: Bacteriophage pRNA
Chain M: Bacteriophage pRNA
Chain N: Bacteriophage pRNA
Chain O: Bacteriophage pRNA

1WAN

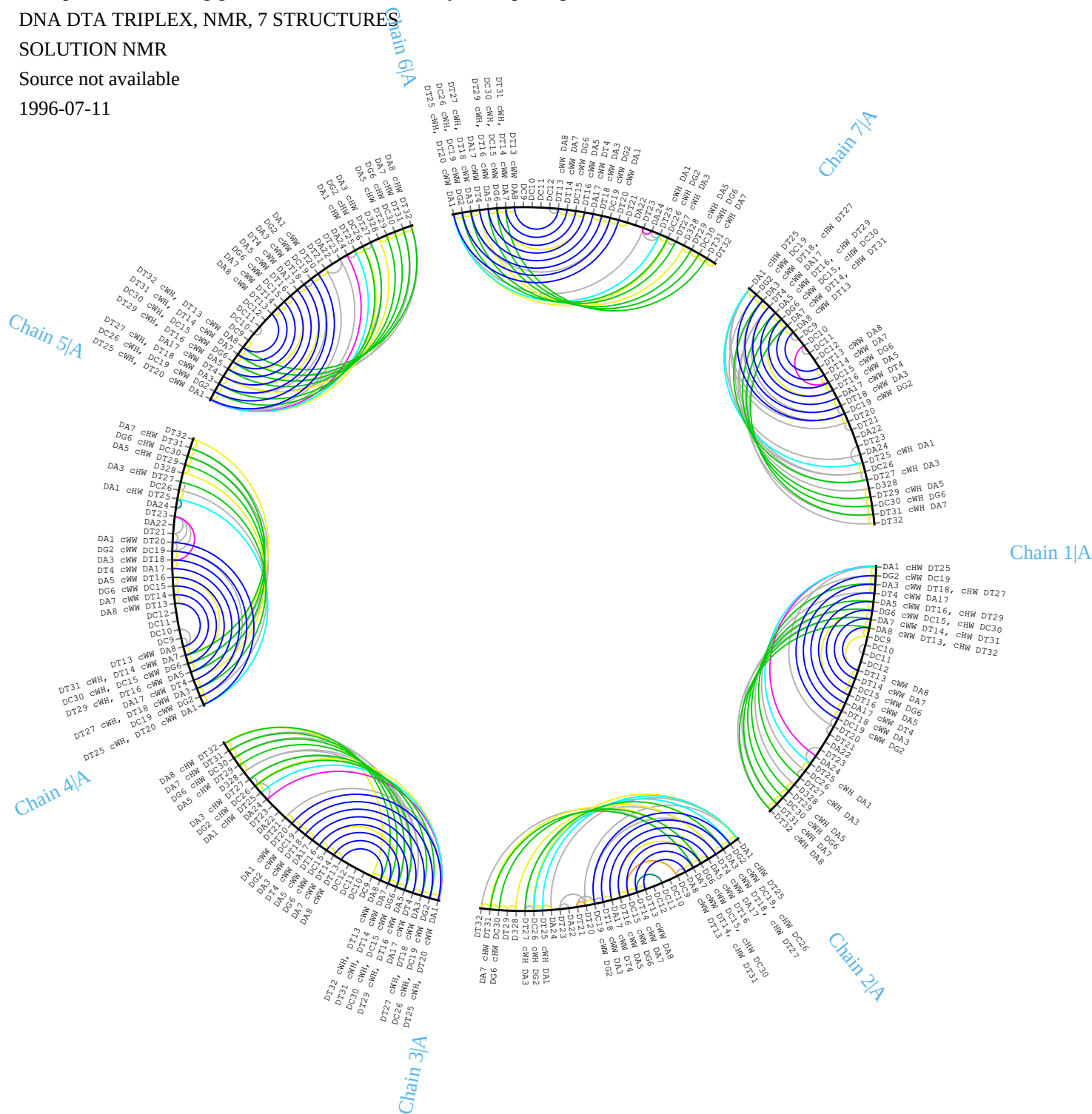
This NMR structure shows 7 models of a DNA triple helix, with nested Watson-Crick basepairs shown in blue and triple helix pairs with crossing green arcs. There is variability in the pairing across the models.

DNA DTA TRIPLEX, NMR, 7 STRUCTURES

SOLUTION NMR

Source not available

1996-07-11



- 53 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 0 long-range Watson-Crick basepairs
- 8 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 33 long-range non-Watson-Crick basepairs
- 141 base stacking interactions
- 6 base-phosphate interactions
- 2 base-ribose interactions
- 0 sugar-ribose interactions
- 2 oxygen stacking interactions
- 75 near basepairs or other interactions

Chain A: DNA (5'-D(*AP*GP*AP*TP*AP*GP*AP*AP*CP*CP*CP*CP*TP*TP*CP*TP*AP*TP*CP*TP*TP*AP*TP*AP*TP*CP*TP*(D3)P*TP*CP*TP*T)-3')

3.1A

Chain B: RNA (5'-R(*GP*GP*CP*UP*CP*GP*GP*CP*GP*GP*CP*GP*GP*A)-3')

4V9O_assembly_4,2

Structure 4V9O has four complete ribosomes in assemblies 1, 2, 3, 4. Here we see assembly 4 followed by assembly 2.

Note the single base-phosphate interaction between the assemblies. Large page size to facilitate zooming.

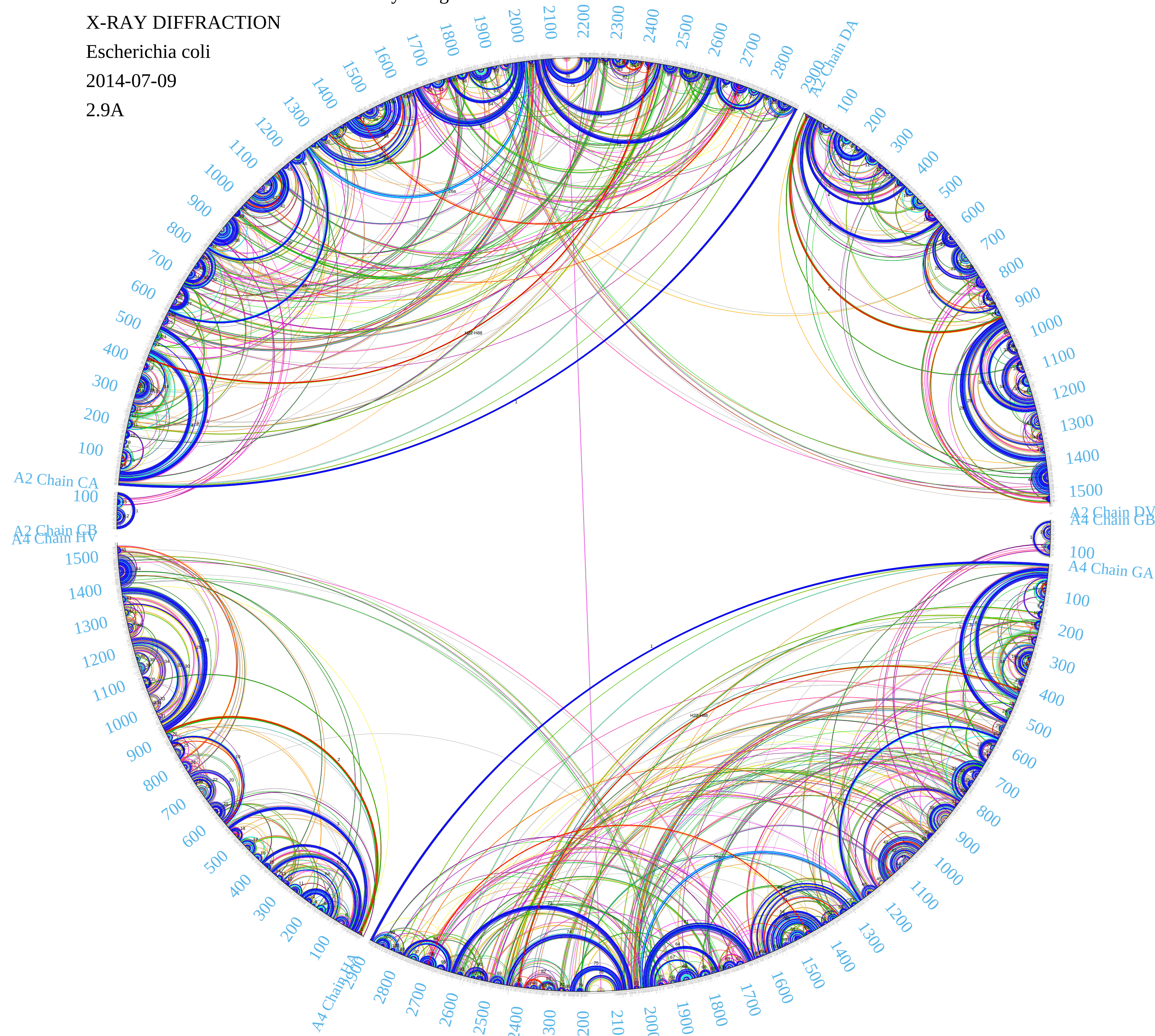
Control of ribosomal subunit rotation by elongation factor G

X-RAY DIFFRACTION

Escherichia coli

2014-07-09

2.9A



- 2266 nested Watson-Crick basepairs (AU, GC, GU cWW)

87 long-range Watson-Crick basepairs

659 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)

403 long-range non-Watson-Crick basepairs

9099 base stacking interactions

2047 base-phosphate interactions

2204 base-ribose interactions

965 sugar-ribose interactions

894 oxygen stacking interactions

3389 near basepairs or other interactions
- Chain GB: 5S ribosomal RNA

Chain GA: Large subunit ribosomal RNA

Chain HA: Small subunit ribosomal RNA

Chain HV: Elongation factor G

Chain CB: 5S ribosomal RNA

Chain CA: Large subunit ribosomal RNA

Chain DA: Small subunit ribosomal RNA

Chain DV: Elongation factor G

9EVS_1

Plant mitochondrial large subunit ribosomal RNA, with helices numbered.

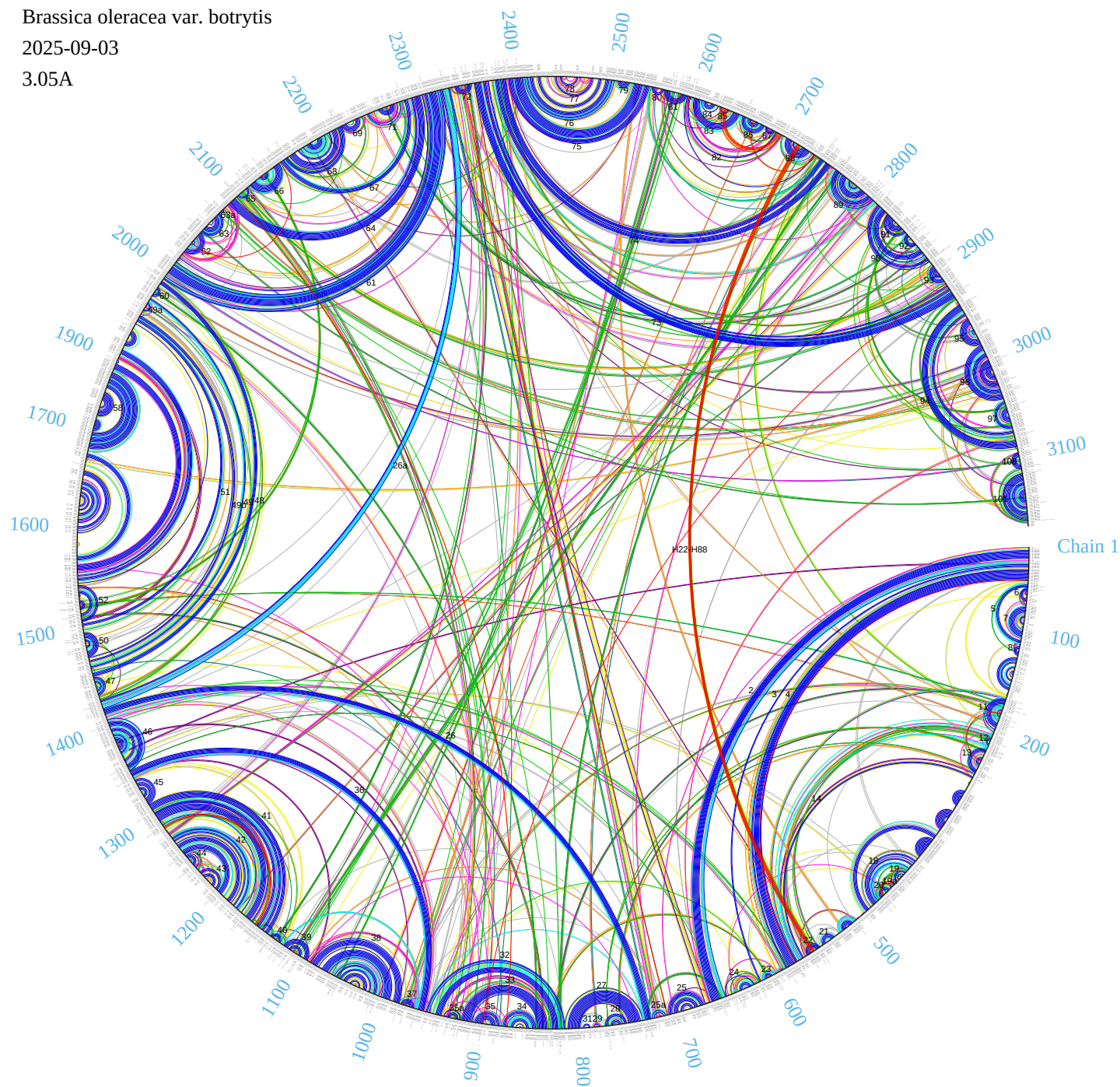
Structure of the flowering plant mitoribosome with P-site tRNA

ELECTRON MICROSCOPY

Brassica oleracea var. botrytis

2025-09-03

3.05A



- 745 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 22 long-range Watson-Crick basepairs
- 243 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 118 long-range non-Watson-Crick basepairs
- 2904 base stacking interactions
- 390 base-phosphate interactions
- 459 base-ribose interactions
- 202 sugar-ribose interactions
- 182 oxygen stacking interactions
- 888 near basepairs or other interactions

Chain 1: Large subunit ribosomal RNA

9I05

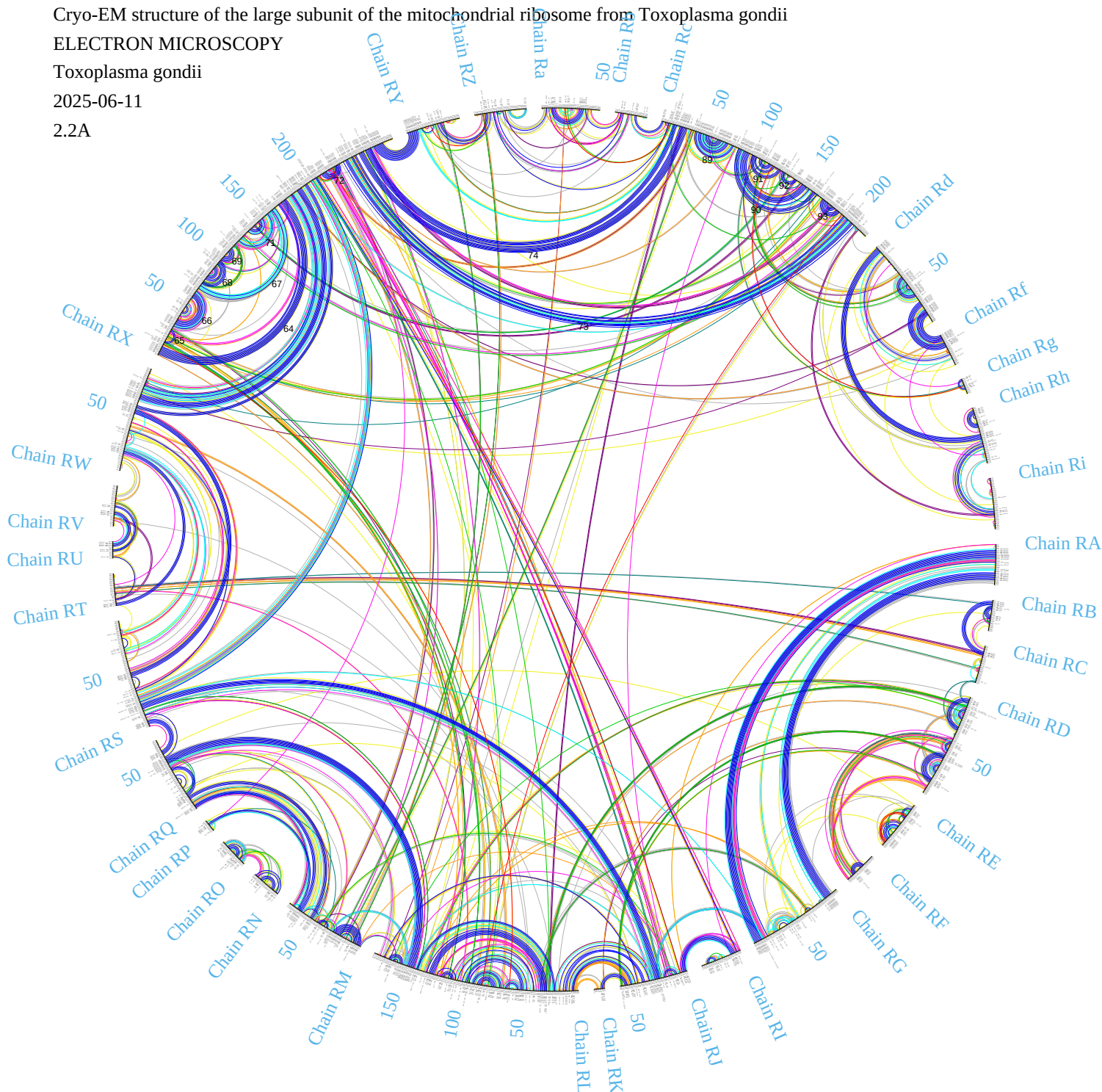
Toxoplasma gondii mitochondrial ribosomal large subunit in 32 chains. Note the interaction types which bind the chains to one another. Helix numbering is not available for short matches to Rfam. Compare to the plant mitochondrial large subunit above. Cryo-EM structure of the large subunit of the mitochondrial ribosome from Toxoplasma gondii

ELECTRON MICROSCOPY

Toxoplasma gondii

2025-06-11

2.2A



- 328 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 9 long-range Watson-Crick basepairs
- 143 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 46 long-range non-Watson-Crick basepairs
- 1593 base stacking interactions
- 172 base-phosphate interactions
- 212 base-ribose interactions
- 73 sugar-ribose interactions
- 78 oxygen stacking interactions
- 596 near basepairs or other interactions

Chain RA: LSU-2
Chain RB: LSU-3
Chain RC: LSU-4
Chain RD: LSU-5
Chain RE: LSU-6
Chain RF: LSU-7
Chain RG: LSU-8
Chain RH: LSU-9
Chain RI: LSU-10
Chain RJ: LSU-11
Chain RK: LSU-12
Chain RL: LSU-13
Chain RM: LSU-14
Chain RN: LSU-15
Chain RO: LSU-16
Chain RP: LSU-17
Chain RQ: LSU-18
Chain RS: LSU-19
Chain RT: LSU-20
Chain RU: LSU-21
Chain RV: LSU-22
Chain RW: LSU-23
Chain RX: Large subunit ribosomal RNA
Chain RY: LSU-24
Chain RZ: LSU-25
Chain Ra: LSU-26
Chain Rb: LSU-27
Chain Rc: Large subunit ribosomal RNA
Chain Rd: LSU-28
Chain Re: LSU-29
Chain Rf: LSU-30
Chain Rg: LSU-31
Chain Rh: LSU-32
Chain Ri: LSU-32

Two assemblies of a DNA double helix with DG triphosphate ligand associated with a protein chain. In assembly 2, DGT makes a Watson-Crick basepair, in assembly 1 just a stacking interaction. The ligand chain appears in a natural location between DNA chains. Dpo4 extension ternary complex with a C base opposite the 2-aminofluorene-guanine [AF]G lesion

Source not available

2.7A



- 27 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 0 long-range Watson-Crick basepairs
- 0 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 0 long-range non-Watson-Crick basepairs
- 63 base stacking interactions
- 1 base-phosphate interactions
- 0 base-ribose interactions
- 0 sugar-ribose interactions
- 0 oxygen stacking interactions
- 7 near basepairs or other interactions

Chain D: 5'-D(*GP*TP*TP*GP*GP*AP*TP*GP*GP*TP*AP*GP*(DOC))-3'

Chain A: DNA polymerase IV

Chain E: 5'-D(*CP*CP*TP*A*AP*CP*GP*CP*TP*AP*CP*CP*AP*TP*CP*CP*AP*AP*CP*C)-3'

Chain H: 5'-D(*GP*TP*TP*GP*GP*AP*TP*GP*GP*TP*AP*GP*(DOC))-3'

Chain B: DNA polymerase IV

Chain J: 5'-D(*CP*CP*TP*A*AP*CP*GP*CP*TP*AP*CP*CP*AP*TP*CP*CP*AP*AP*CP*C)-3'

1ENE

Chain A interacts with a symmetry operated counterpart of Chain A. Short yellow arcs indicate stacking between adjacent nucleotides; long yellow arcs indicate cross-strand stacking.

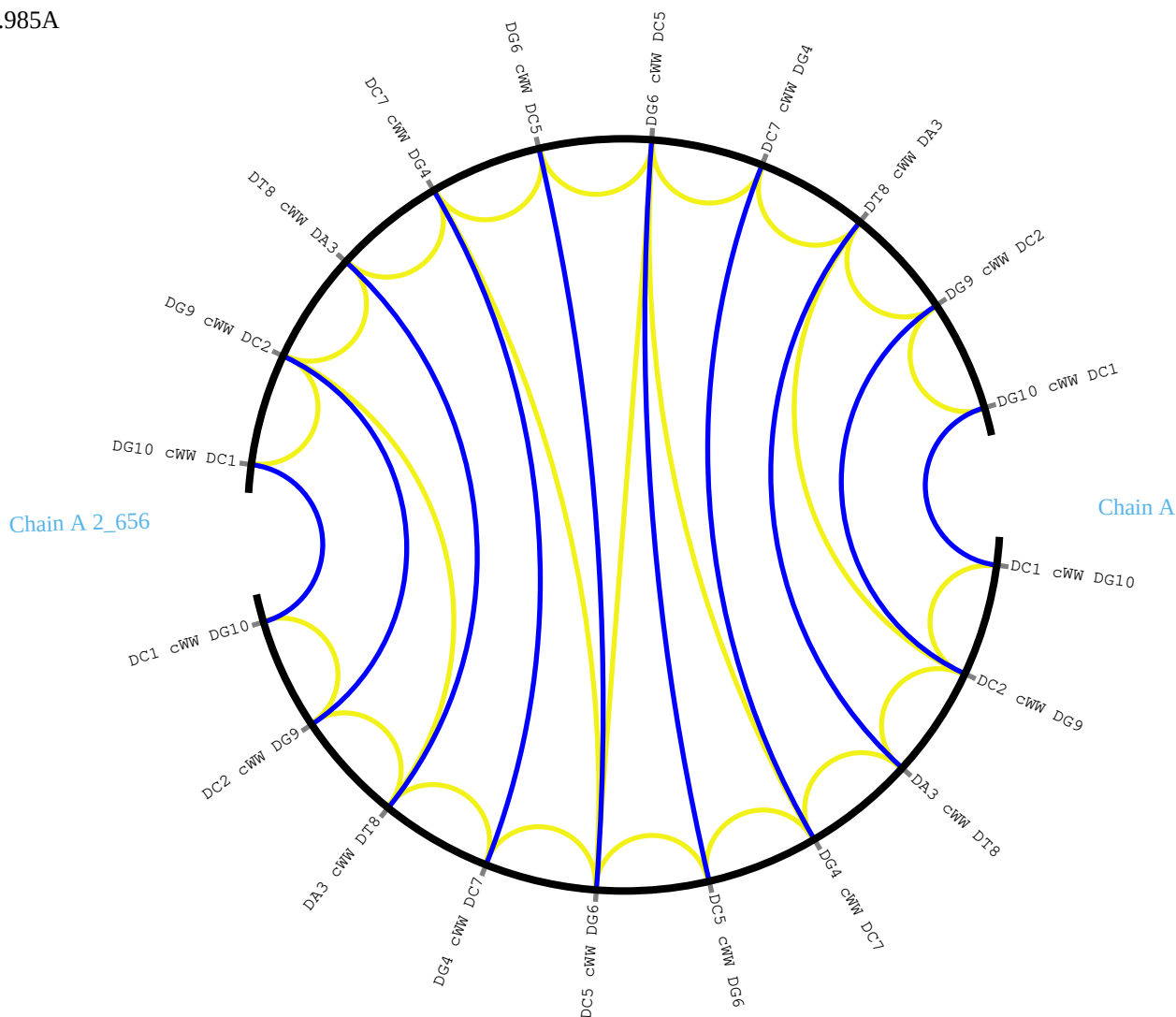
1A CRYSTAL STRUCTURES OF B-DNA REVEAL SEQUENCE-SPECIFIC BINDING AND GROOVE-SPECIFIC BENDING OF DNA BY MAGNESIUM AND CALCIUM.

X-RAY DIFFRACTION

Source not available

2000-09-25

0.985A



- 10 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 0 long-range Watson-Crick basepairs
- 0 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 0 long-range non-Watson-Crick basepairs
- 23 base stacking interactions
- 0 base-phosphate interactions
- 0 base-ribose interactions
- 0 sugar-ribose interactions
- 0 oxygen stacking interactions
- 0 near basepairs or other interactions

Chain A: DNA (5'-D(*CP*CP*AP*GP*CP*GP*CP*TP*GP*G)-3')

6HU6_A_B_F_E

Rather than ordering alphabetically, chain order is specified to keep Watson-Crick paired chains adjacent to one another.

At the break between chains A and B, helices are stacked on each other with an offset that results in O4' atoms stacking on bases.

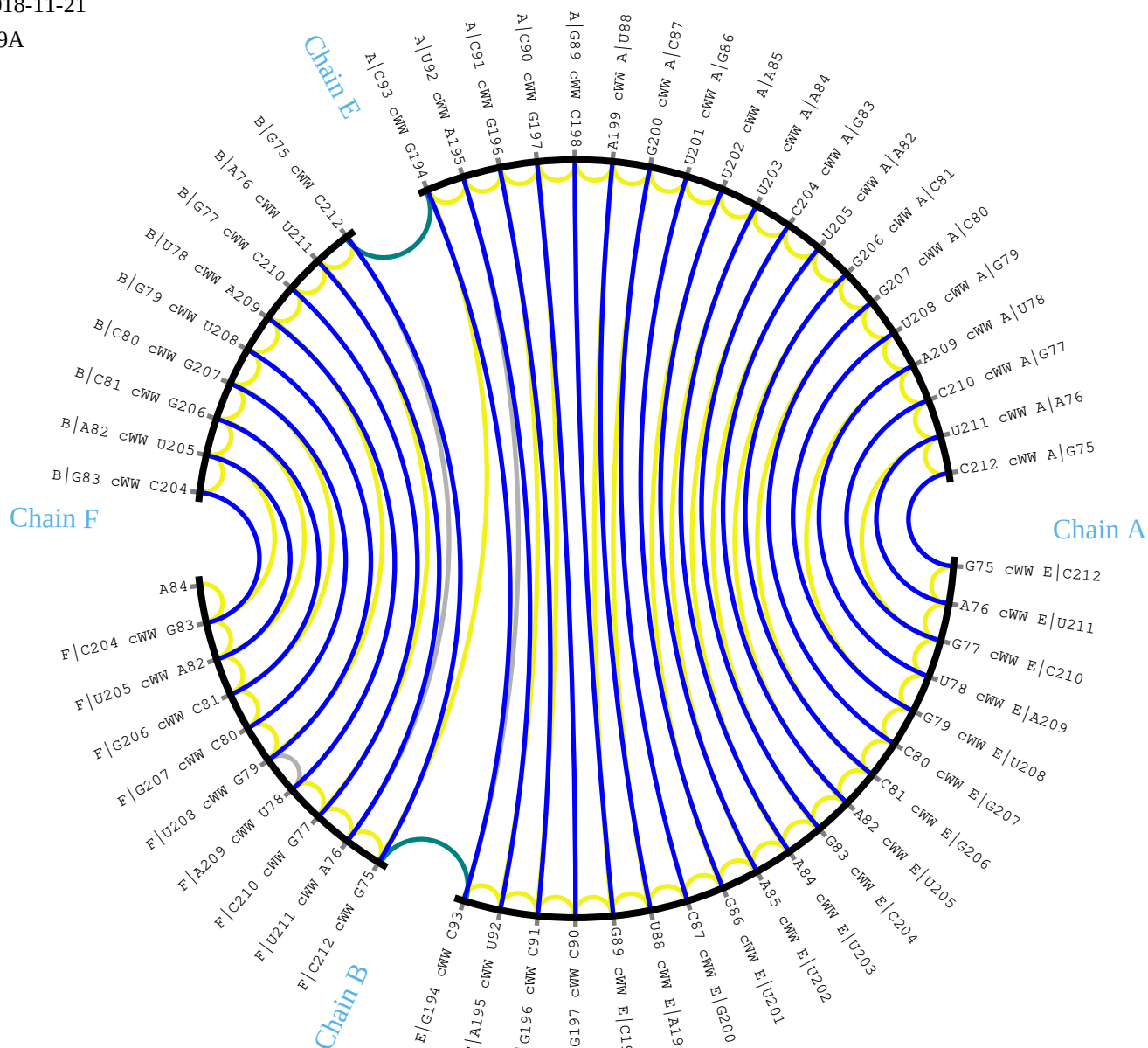
Structure of ARF1 RNA

X-RAY DIFFRACTION

Homo sapiens

2018-11-21

1.9A



- 28 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 0 long-range Watson-Crick basepairs
- 0 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 0 long-range non-Watson-Crick basepairs
- 70 base stacking interactions
- 0 base-phosphate interactions
- 0 base-ribose interactions
- 0 sugar-ribose interactions
- 2 oxygen stacking interactions
- 3 near basepairs or other interactions

Chain A: RNA (19-MER)

Chain B: RNA (5'-R(*GP*AP*GP*UP*GP*CP*CP*AP*GP*A)-3')

Chain F: RNA (5'-R(*CP*UP*GP*GP*UP*AP*CP*UP*C)-3')

Chain E: RNA (19-MER)

2IZ8

Viral capsid. Assembly 1 has 60 repetitions of the same unit using symmetry operators. Assembly 3 has five units. Assembly 4 has six units.

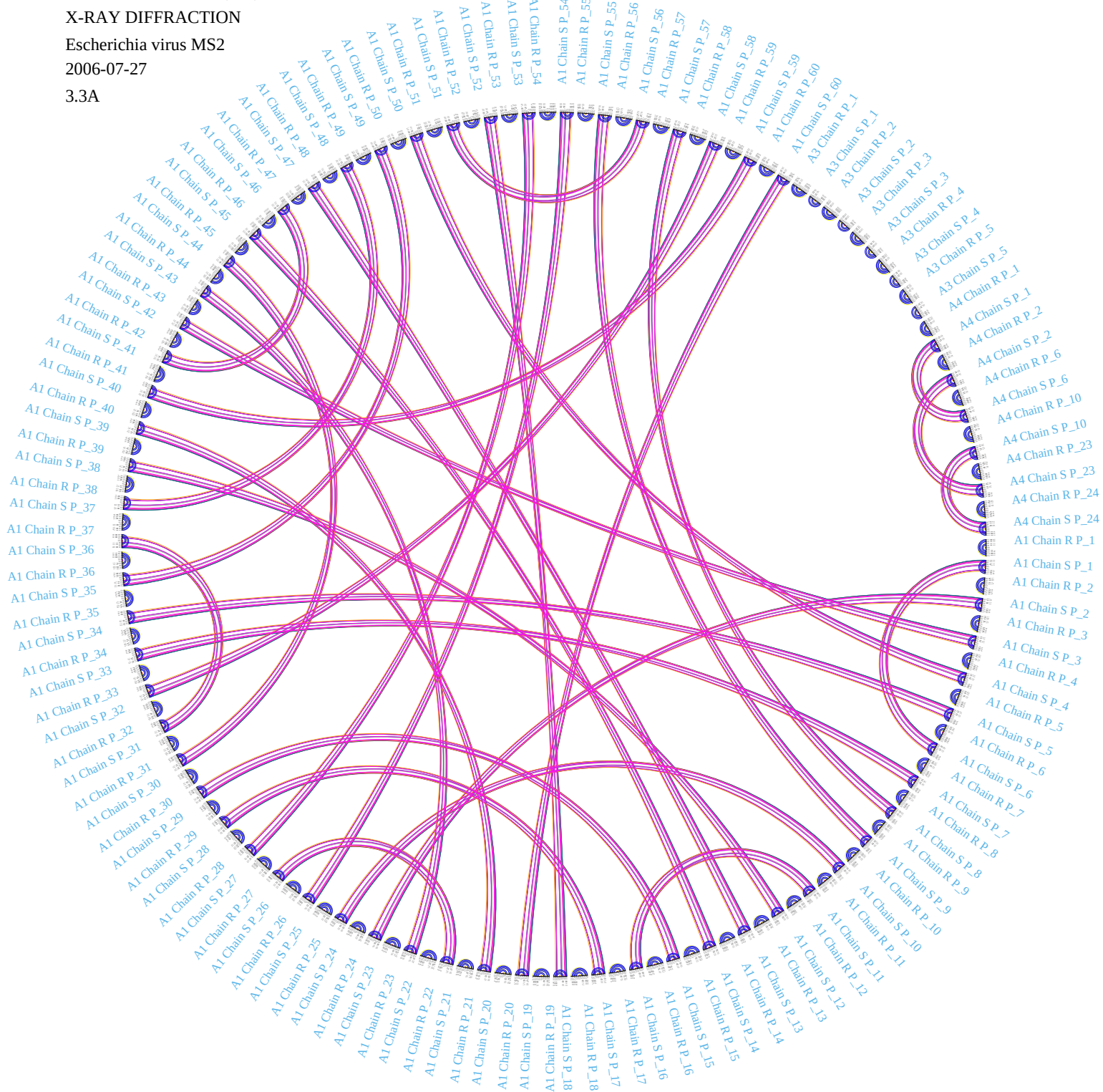
MS2-RNA HAIRPIN (C-7) COMPLEX

X-RAY DIFFRACTION

Escherichia virus MS2

2006-07-27

3.3A



- 639 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 0 long-range Watson-Crick basepairs
- 0 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 0 long-range non-Watson-Crick basepairs
- 1846 base stacking interactions
- 274 base-phosphate interactions
- 66 base-ribose interactions
- 66 sugar-ribose interactions
- 132 oxygen stacking interactions
- 274 near basepairs or other interactions

Chain R: RNA, (5'-R(*AP*CP*AP*UP*GP*AP*GP*GP*CP*UP*CP* AP*CP*CP*CP*AP*UP*GP*U)-3')

Chain S: RNA, (5'-R(*AP*CP*AP*UP*GP*AP*GP*GP*CP*UP*CP* AP*CP*CP*CP*AP*UP*GP*U)-3')

9HIO

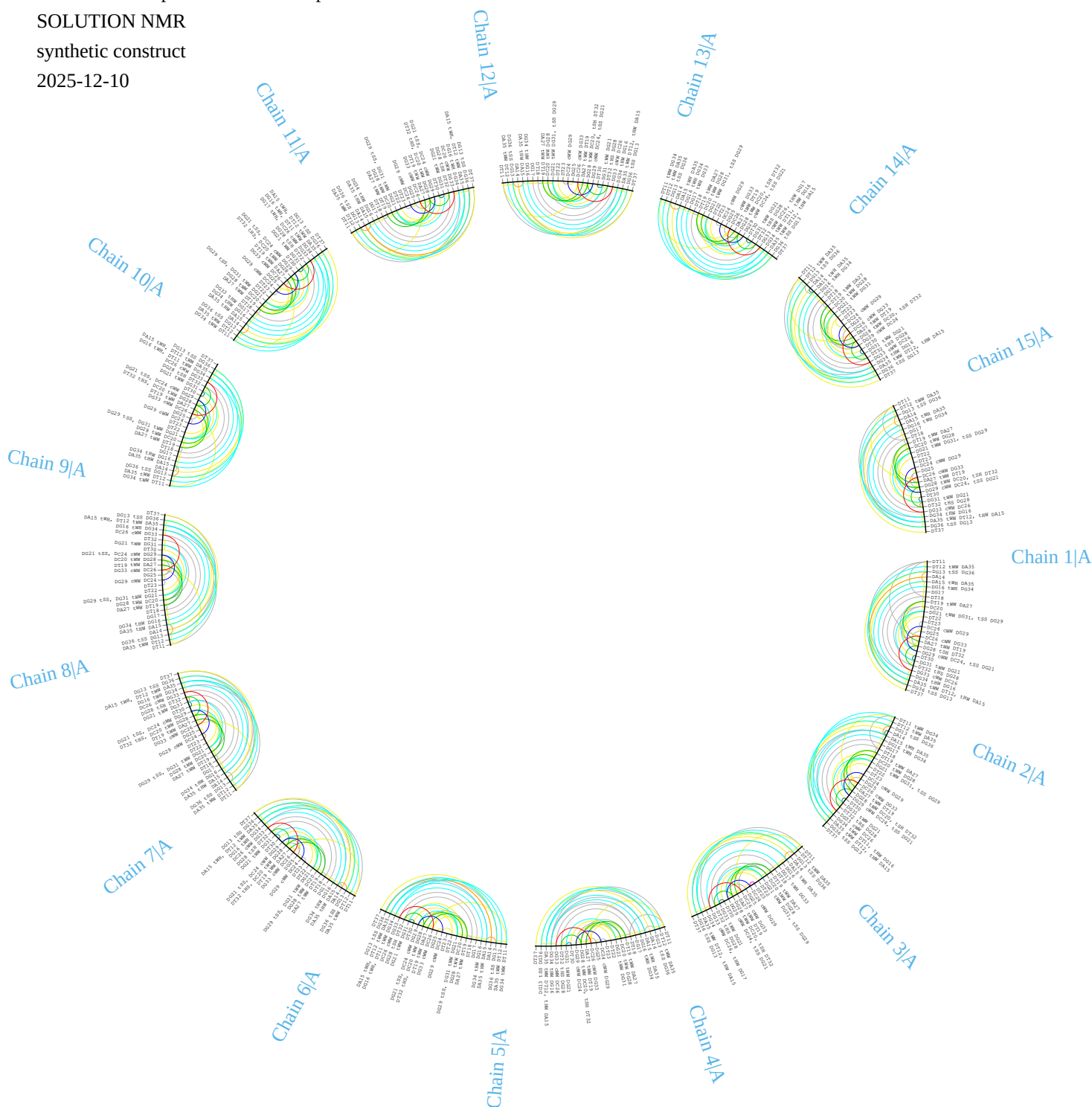
NMR structure of a DNA molecule with many non-Watson-Crick basepairs and base-backbone interactions.

RKEC1 DNA aptamer bound to dopamine

SOLUTION NMR

synthetic construct

2025-12-10



- 15 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 15 long-range Watson-Crick basepairs
- 95 nested non-Watson-Crick basepairs (Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 43 long-range non-Watson-Crick basepairs
- 344 base stacking interactions
- 1 base-phosphate interactions
- 19 base-ribose interactions
- 0 sugar-ribose interactions
- 16 oxygen stacking interactions
- 128 near basepairs or other interactions

Chain A: RKEC1 DNA (27-MER)

7EI6

Hybrid chain, first 20 nucleotides are RNA, last 15 are DNA. Note the Watson-Crick helices with RNA-DNA basepairs, the internal loop with two non-Watson-Crick basepairs, and the GAAA hairpin with three base-backbone interactions characteristic of a GNRA hairpin.

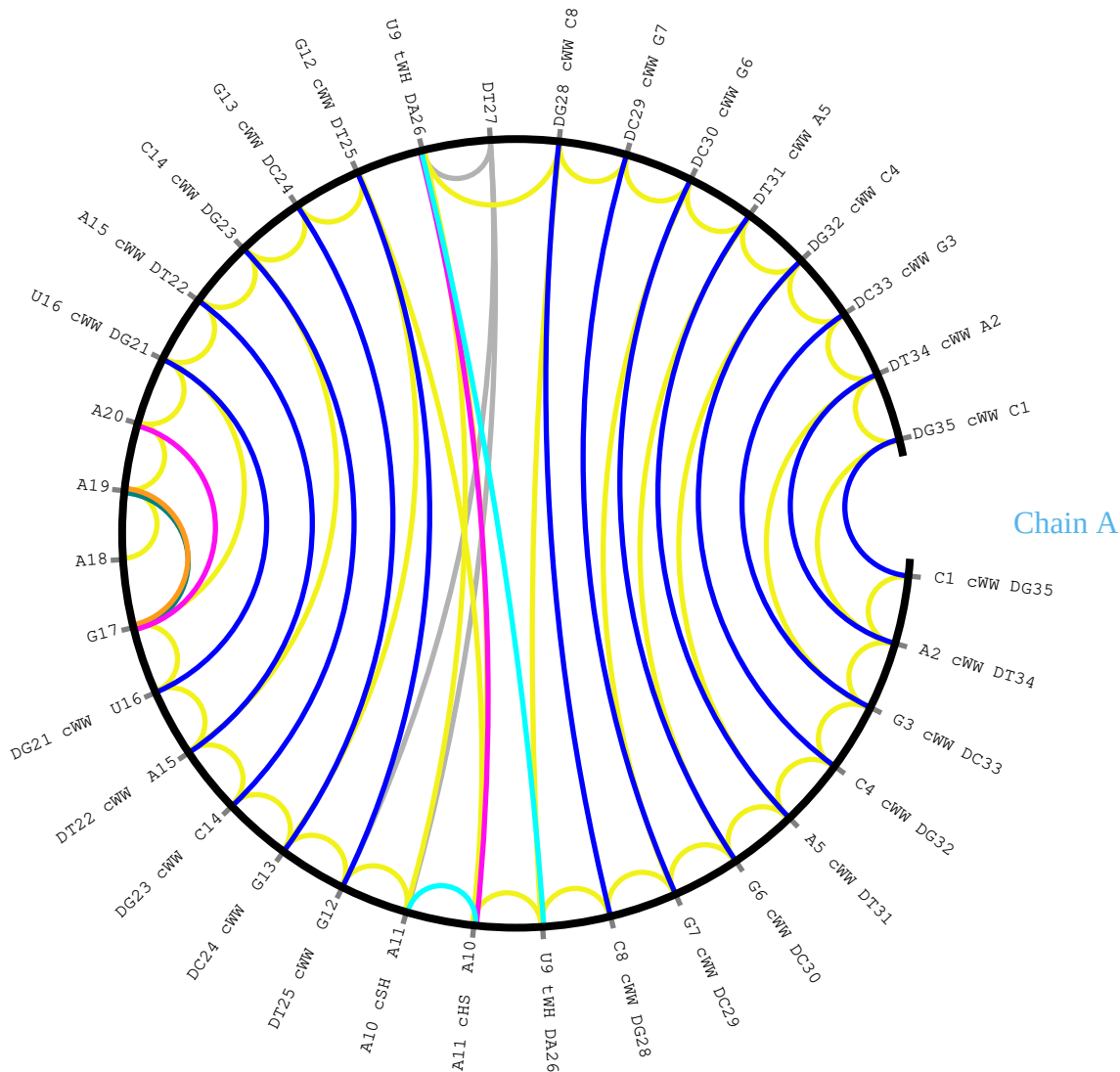
GNRA tetraloop receptor motif composed of RNA and DNA

X-RAY DIFFRACTION

synthetic construct

2022-03-30

2.9A



- 13 nested Watson-Crick basepairs (AU, GC, GU cWW)
- 0 long-range Watson-Crick basepairs
- 2 nested non-Watson-Crick basepairs
(Includes AA cWW, AG cWW, UU cWW, AG tHS, etc.)
- 0 long-range non-Watson-Crick basepairs
- 42 base stacking interactions
- 2 base-phosphate interactions
- 1 base-ribose interactions
- 0 sugar-ribose interactions
- 1 oxygen stacking interactions
- 3 near basepairs or other interactions

Chain A: DNA/RNA (35-MER)